FINITE TYPE SUBMODULES EXTEND ACROSS QUASI-COMPACT OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2763-2778, following the statement at 2751-2761.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Pushforward of quasi-coherent modules. Morphisms of Spaces, Lemma spaces-morphisms-lemma-pushforward (cross-chapter, cited at ~2764-2766). j_*G and j_*F|_U are quasi-coherent on X (j quasi-compact).
- Directed colimit of finite type submodules. lemma-directed-colimit-finite-type (same chapter, card L26345, cited at ~2770-2771). Find finite type H' in H with H'|_U = G.

Source reference. Original lemma 0829: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2519–2519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M9: focused-seeds/geometry/S9-spaces-limits-extend/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $U \subset X$ be a quasi-compact open. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $\mathcal{G} \subset \mathcal{F}|_U$ be a quasi-coherent $\mathcal{O}_U$-submodule which is of finite type. Then there exists a quasi-coherent submodule $\mathcal{G}' \subset \mathcal{F}$ which is of finite type such that $\mathcal{G}'|_U = \mathcal{G}$.*

Proof. Denote $j : U \to X$ the inclusion morphism. As X is quasi-separated and U quasi-compact, the morphism j is quasi-compact. Hence $j_*\mathcal{G} \subset j_*\mathcal{F}|_U$ are quasi-coherent modules on X (Morphisms of Spaces, Lemma 03M9). Let $\mathcal{H} = \mathrm{Ker}(j_*\mathcal{G} \oplus \mathcal{F} \to j_*\mathcal{F}|_U)$. Then $\mathcal{H}|_U = \mathcal{G}$. By Lemma 0829 we can find a finite type quasi-coherent submodule $\mathcal{H}' \subset \mathcal{H}$ such that $\mathcal{H}'|_U = \mathcal{H}|_U = \mathcal{G}$. Set $\mathcal{G}' = \mathrm{Im}(\mathcal{H}' \to \mathcal{F})$ to conclude. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.